STATEMENT OF WORK

VLS DECK AND HATCH ASSEMBLY

N63394-26-R-4002

1. BACKGROUND

The Naval Surface Warfare Center, Port Hueneme Division (NSWC PHD) is required to procure Deck and Hatch Assembly (Assy) for MK 41 Vertical Launching System (VLS). Deck and Hatch Assembly is the upper part of a MK 41 VLS module which contains eight (8) Cell Hatches and one (1) Uptake Hatch along with the mechanical parts necessary to open them. It also includes the Module Deluge System to cool any Canisters in the Module in the event of a casualty. It contains the framework necessary to attach it to the lower part of the module which must be precisely aligned. In the center of the Deck and Hatch Assy is the Upper Uptake which must be properly sealed to prevent missile exhaust from escaping.

1.1 SCOPE

The Contractor shall be responsible for the satisfactory and timely performance of all tasks defined in the Statement of Work (SOW). This SOW, with its supporting attachments and referenced documentation, describes the requirement for the manufacturer, assembly, test and delivery of VLS Deck and Hatch Assy.

1.2 PROGRAM AUTHORITY

Naval Surface Warfare Center, Port Hueneme Division (NSWC PHD) is the program authority for this contract.

2. APPLICABLE DOCUMENTS

The following documents, of the issue indicated, form a part of this SOW to the extent specified herein, including all applicable terms, conditions, clauses, attachments and exhibits. The exact revisions of the documents are not cited in the text of this SOW. The Contractor shall apply the latest version of the documents in effect at the time of award of the contract. In the event of a conflict between the documents referenced herein and the contents of this SOW, the SOW shall be considered the superseding requirement.

2.1 Government Documents

Document Number	Document Title
MIL-STD-109C	Quality Assurance Terms and Definitions
MIL-STD-1686C	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, I Assemblies and Equipment (Excluding

	Electrically Initiated Explosive Devices)
MIL-STD-1689	Fabrication, Welding, and Inspection of Ship Structure
WS 20130	Production Environmental Test Requirements for VLS
WS20350	Weapon Specification, Vertical Launching System (VLS), MK 41
TL130-AD-PLN-010-VLS	Configuration Management Program Plan for the MK 41 VLS, Volume 1, Equipment (Hardware) Configuration Management
MK 41 VLS Standing Instruction 94-1F, Attachment 6	Marking for technical data
70107534	Deck and Hatch Assembly

2.2 Industry Documents

Document Number	Document Title
ANSI/ISO/ASQC Q9001	Quality Systems - Model for Quality Assurance in Design, Development, Production, Installation and Servicing (Latest Version)
ANSI/EIA-649	National Consensus Standard for Configuration Management
ANSI/ASQC Q9004	Quality System Elements
ISO 9001	Quality Program Requirements
SAE AS5553	Counterfeit Electronic Parts: Avoidance, Detection, Mitigation, and Disposition
SAE AS6174	Counterfeit Material: Assuring Acquisition of Authentic and Conforming Material
SAE AS9102 Rev A	Aerospace First Article Inspection
ISO 14000	Environmental Management Systems Standards

3. REQUIREMENTS

The Government requires manufacturing of VLS Deck and Hatch Assy in accordance with the documents specified in SOW and the technical data package (TDP). All Deck and Hatch Assy shall meet all form, fit, and function requirements in accordance with the TDP and associated specifications.

The Contractor shall fabricate, assemble, test, and deliver VLS Deck and Hatch Assy in accordance with technical drawings, and specifications identified in Section C and in accordance with the contents of this SOW. All Deck and Hatch Assy shall meet all form, fit, and function requirements in accordance with the TDP and associated specifications.

3.1 First Article Inspection

The Contractor shall perform First Article Inspection (FAI) of all assemblies, subassemblies, components and detail parts including castings and forgings in accordance with SAE Aerospace Standard AS9102 Rev A. The purpose of FAI is to provide objective evidence that all TDP and specification requirements are correctly understood, accounted for, verified, and recorded. An

FAI is not required for standard catalog items, commercial-off-the-shelf items, raw materials or items procured in accordance with source control drawings.

The Contractor shall submit all FAI requirement waivers to the Government for approval. The waiver shall identify, at a minimum, the part description, part number, manufacturer, date last built, location last built, FAI report summary (if applicable), FAI discrepancies and corrective actions (if applicable), and waiver justification.

The Government reserves the right to witness all FAI activities. The Contractor shall submit a comprehensive schedule of FAI events that includes date, time, expected duration and location to the Government at least 20 working days prior to the first event, in accordance with CDRL A001.

The Contractor shall generate and submit via digital means to the Government FAI reports in accordance with AS9102 and CDRL A002. The Contractor shall retain the FAI reports for at least seven years. The Government reserves the right to examine all FAI reports.

All deficiencies discovered during the FAIs that affect form, fit, function or performance shall be corrected at Contractor’s expense prior to unit delivery to the Government.

3.2 Technical Data Reviews and Quality Audits

During the performance of the contract, the Government has the right to perform technical data reviews and quality audits consisting of evaluation of records, processes and products to verify the Contractor's compliance with the applicable quality programs. The Government audits may include evaluation of effectiveness in implementing the Contractor's and Subcontractors' or suppliers' audit programs to promote prevention of defects, which includes schedule of audits, evaluation of operations, notification of required corrective action with follow-up and means of notification to top management concerning audit results.

3.3 Fit Check Tools

The Contractor shall be responsible for fabricating or procuring, maintaining, and calibrating all tools required to perform Module Assembly canister/cell interface fit checks, as specified in Canister/Cell Interface Performance Specification (CIPFS), WS 20350, paragraph 4.2.5.1.

Prior to initial use, the Contractor shall perform a First Article Inspection (FAI) of each fit check tool in accordance with SAE Aerospace Standard AS9102. Each FAI may be witnessed, at the Government’s discretion, by a designated Government representative.

The Contractor shall calibrate each tool prior to initial use for Module Assembly canister/cell interface fit checks and shall re-calibrate each tool annually thereafter.

3.4 Follow-On Physical Configuration Audit

The Physical Configuration Audit (PCA) purpose is to verify the as-built configuration matches its technical documentation and validate supporting the processes. The PCAs shall be conducted in accordance with Section 7 of TL130-AD-PLN-010-VLS, Configuration Management Program Plan for the MK 41 VLS.

During the performance of the contract, the Government has the right to perform a Physical Configuration Audit (PCA). The PCA may include an evaluation of the effectiveness of the Contractor, Subcontractor and/or supplier audit programs to promote prevention of defects. This includes scheduling of audit, evaluation of operations, and notification of required corrective action with follow-up and means of notification to top management concerning audit results.

One PCA shall be conducted over the course of the contract on the VLS Deck and Hatch Assy. The VLS Deck and Hatch Assy shall be selected by the Government and the Contractor from the production run.

The Contractor shall provide qualified engineering, technical, manufacturing, administrative, management and other support as necessary for the PCA. The Contractor shall provide all tools, measuring devices and other equipment required for equipment disassembly, dimensional and tolerance verification and performance verification. The Contractor shall provide an area to conduct PCA that is well illuminated, temperate and low noise level. The PCA work area shall contain worktables, print copies of all pertinent drawings, telephone and easy access to a copy machine.

During the PCA, the Contractor shall provide, at a minimum, individual parts that make up the configuration item, completed assemblies, an as-built list delineating all Engineering Change Proposal (ECP) (either requested or approved), deviations and waivers, hardcopies of production specifications and technical drawings, complete shortage list, manufacturing instructions, acceptance test procedures and data, purchase order documents and associated certificates of conformance from suppliers or appropriate documents indicating compliance with material or drawing requirements in accordance with CDRL A003.

The PCA will be conducted at the Contractor's, Subcontractors' and suppliers' facilities by the Government team. The Government will notify the Contractor seven (7) to 14 business days prior to performance of the PCA. This notice will include identification of assemblies, subassemblies and/or piece parts to be audited. The PCA will be approximately one to two weeks in duration.

Following conduct of the PCA, the Contractor shall generate the necessary documentation to resolve all audit findings and provide these to the Government audit team within the timeframe specified by the Government in accordance with CDRL A004. The Government audit team will evaluate the documentation to assess corrective action(s) taken to close any and all audit finding(s). Necessary documentation may include, but not be limited to, revised manufacturing processes, ECPs, notice of revisions, SCN and revised TDP drawings.

Government and Contractor pre-and post-PCA activities are described in Sections 7.4 through 7.7 of the VLS Configuration Management Plan TL130-AD-PLN-010-VLS.

All deficiencies discovered during the PCA that affect form, fit, function or performance shall be corrected at Contractor's expense prior to the unit delivery to the Government.

3.5 Government Inspection

The Contractor shall provide technical, administrative, and management support for Government source inspection for purposes of inspection and acceptance of supplies and services provided by the Contractor, Subcontractor and suppliers. Government source inspection will be conducted at the convenience of the Government.

3.6 Production Environmental Tests

Production environmental testing as specified by the TDP or Production Environmental Test Requirements for VLS Specification WS 20130 shall be performed.

3.7 Technical Data Rights

The Government will maintain the Government-owned TDP. The Contractor shall submit proposed changes to the TDP to the Government. The Government will retain unlimited rights to all technical data, versions of and changes to all drawings, specifications and other documents for the VLS components produced under this contract.

3.8 Marking for Technical Data

The Contractor shall mark all technical data in accordance with MK 41 VLS Standing Instruction 94-1F, Attachment 6. This instruction is applicable to new documents, revised documents or documents from file in response to a request for copy.

3.9 Manufacturing Processes

The Contractor shall maintain and control manufacturing processes to fabricate, assemble, inspect, test, and deliver VLS Deck and Hatch Assy as described in the SOW. The processes shall cover all phases from the ordering of raw materials to Government acceptance of the finished product. All hardware components shall be manufactured in accordance with the drawings and specifications referenced in the TDP and this SOW.

Documentation and records of the manufacturing processes and products, such as test reports, material certification, QA records, etc. shall be maintained and provided to the Government upon request.

3.10 Configuration Management

The Contractor shall maintain a configuration management program, which shall provide for the administrative and functional systems necessary for configuration identification, control, status accounting and reporting, to ensure configuration identity with the VLS Deck and Hatch Assy produced by the Contractor. The Contractor shall maintain a Contractor approved Configuration Management Plan that complies with ANSI/EIA-649 2011.

Notwithstanding ANSI/EIA-649 2011, the Contractor's configuration management program shall comply with the VLS Configuration Management Plans, TL130-AD-PLN-010-VLS, and shall comply with the following:

3.10.1 Configuration Identification

The Contractor shall maintain identification of the product baselines throughout fabrication, testing, and delivery. The Contractor shall ensure that equipment is assigned identifying numbers, designators, and serial numbers in accordance with the TDP.

Serialized items shall be traceable to source of manufacture and production contract year. This information may be incorporated as a formal part of the Contractor's serialization system or otherwise be annotated on the affected item.

The request for serial number assignment shall contain the following minimum information:

- (a) Officially assigned item name
- (b) Officially assigned type designation
- (c) Officially assigned model number
- (d) Top drawing number and list of drawings or parts list
- (e) Exact quantity to be delivered under the contract including pre-production samples and spares required by the contract.
- (f) Contract number
- (g) National Stock Number
- (h) NAVSEA cognizant office code

The Contractor shall submit a configuration definition data package describing the detailed configuration of each end item as delivered to the designated destination in accordance with CDRL A004. The Contractor shall maintain a system to monitor engineering release and correlation of manufactured products.

3.10.2 Military Specifications and Standards

The VLS Deck and Hatch Assy TDPs, reference and cite cancelled military standards and specifications. The Government has no plan to replace any such standard or specification with a commercial version. The military standard or specification of the latest version in effect at time of cancellation shall be considered applicable to this procurement.

3.10.3 Qualification of Substitute Sources of Supply

The TDP contains source control or vendor item drawings that specify approved, or suggested sources of supply. The Contractor may qualify other sources for these items. Qualification of new sources selected by the Contractor shall be at the Contractor's expense.

The Contractor shall submit all proposed changes to approved, or suggested sources of supply as indicated in the source control or vendor item drawings in accordance with SOW. The package of information that is submitted for Government review shall contain all applicable drawings, test plans, procedures and qualification reports. An extended Government review period of 20 business days applies to all proposed changes to approved or suggested sources of supply.

3.10.4 Configuration Status Accounting

The Contractor's configuration status accounting shall be in accordance with ANSI/EIA-649, the Configuration Management Plan for the VLS, Volume 1, Equipment (Hardware) Configuration Management, TL130-AD-PLN- 010-VLS, and the Configuration Management Plan for the VLS, Volume 3, Firmware Configuration Management, and TL130-AD-PLN-030-VLS.

The Contractor shall provide documentation of the exact delivered configuration of each unit. The Contractor, or Subcontractor at the appropriate tier, shall identify the "as-built" production configuration of each NAVSEA nomenclature item by means of permanent markings on the appropriate ORDALT/ Ship Change Installation Procedure (SCIP) label plate on the equipment to identify the specific ORDALTs/SCIPs installed in the equipment being delivered. For Class I ECPs, which are not assigned an ORDALT number, the ORDALT label plate shall be marked with the applicable Class I ECP number as assigned by the Government. For purposes of this requirement, markings may be etched, steel stamped or affixed by other suitable means which results in the permanent identification of the equipment configuration. Ink stamping is not permitted.

3.10.5 Failure/Defect Reporting and Corrective Action System (FACAR)

The Contractor shall have a FACAR system established and maintained for both manufacturing and test processes. Level of assembly for failure reporting shall be consistent with the requirement for recording defect history. FACAR system information required to be recorded or reported by this SOW paragraph shall be submitted in accordance with CDRL A005.

The Contractor shall implement an effective system for evaluation and disposition of supplies and services that exhibit nonconformance, including those furnished by suppliers (Subcontractors and vendors). The Contractor shall ensure that the personnel staffing the FACAR system have the authority to coordinate and implement the necessary failure and problem reporting, analysis, and related corrective action. The Contractor shall ensure that nonconformance data is adequately analyzed, distributed to appropriate organizations and personnel, and that appropriate management decisions are based on this data. Analysis must target root cause corrective action. The Contractor shall take appropriate actions to improve or change processes that do not meet requirements. All use-as- is and repair dispositions shall include a determination of whether or not a change is required to the product database (drawing, specifications, work instructions, tooling, etc.). Determination that such a change is not required shall be recorded and shall be available to the Government for review. Acceptance of the nonconformance corrective action and disposition is the prerogative of the Government and can be revoked whenever the Contractor's system fails to demonstrate effective root cause corrective action or when repeated nonconformance indicate an overall quality system degradation.

3.10.6 Configuration Control

Minor Deviations:

1. The contractor shall submit the deviation to NSWC DD for review & approval.
2. If NSWC DD concurs in classification, the deviation will be boarded at NSWC DD CCB for approval or disapproval.
3. If NSWC DD does not concur in classification, the Contractor shall cancel the minor deviation.

The Contractor shall submit for review and approval all specification changes to NSWC DD for review and approval.

3.11 Testing, Inspection and Acceptance

Testing and inspection of all VLS Deck and Hatch Assy shall be performed at the Contractor's facility in accordance with the applicable drawings, standards and specifications and factory test and inspection plans. The Contractor shall assure that all such testing and inspections required for acceptance are satisfactorily performed.

The Contractor shall prepare specific written inspection and test procedures for each inspection and test operation to be performed by the Contractor. These inspection and test procedures shall be reviewed by the quality assurance organization and maintained current in accordance with the change control system requirements. The inspection and test procedures shall clearly identify the item through part number with revision and nomenclature, and include detail instructions and/or operations to be performed.

The Contractor shall maintain records of inspection results and quantitative test data for each deliverable end item and make them available to the Government upon request.

3.12 Parts Re-Screening

The Contractor is not required to implement the TDP requirements to conduct re-screening of electrical components. Remarking of parts to the re-screened part number will not be required and the existing base part number shall remain intact. Programmable parts shall require re-identification.

3.13 Counterfeit Parts and Materials

The following minimum processes shall be implemented and flowed down by the Contractor to all Subcontractors in order to minimize the risk of use of counterfeit parts in VLS components.

The Contractor shall establish processes to minimize the risk of procuring and using counterfeit parts and materials. The Contractor shall document these processes and provide those documented processes to Government upon request. These processes shall include:

- At a minimum, these processes shall ensure that all components for Deck and Hatch and subassemblies are purchased from Original Equipment Manufacturers (OEM), Original Component Manufacturers (OCM) authorized suppliers or franchised distributors. The term OEM, OCM, authorized supplier and franchised distributor are each defined in SAE AS5553).
- At a minimum, the contractor shall ensure procurement practices and processes to purchase and install components from OEM, OCM, authorized suppliers and franchised distributors are flowed down to subcontractors and suppliers at all tiers.
- The Counterfeit Prevention Plan shall be generated in accordance with Data Item Description (DID) DI-MISC-81832, Counterfeit Prevention Plan, requirements.
- The Counterfeit Prevention Plan, SAE AS5553 and SAE AS6174 requirements shall be flowed down to all subcontractors and suppliers.
- The Contractor shall maximize the use of authentic, originally designed and/or qualified parts.
- The Contractor shall assess potential sources of supply to minimize the risk of receiving counterfeit parts or materials.
- The Contractor shall have purchasing procedures which confirm whether a selected supplier is authorized, as defined in SAE AS5553 and SAE AS6174, for each purchase.
- The Contractor shall define minimum inspection and test requirements for parts being procured and shall ensure that in-house, third-party, and/or supplier inspection and test procedures and facilities comply with these requirements. These minimum inspection and test requirements shall specify appropriate test methods to detect potential counterfeit parts and materials.
- The Contractor shall require a certificate of compliance and supply chain traceability for all electronic part purchases.
- The Contractor shall require a Certificate of Conformance, as defined in SAE AS5553 and SAE AS6174, and supply chain traceability for all electronic part purchases.
- The Contractor shall use government or industry services such as the Government-Industry Data Exchange Program (GIDEP) and other commercially available services to identify part or supplier quality or authenticity problems.

The Contractor shall notify the Government of the occurrence of a confirmed counterfeit part or material and the actions taken to identify, contain, and impound all product from the lot, within seven (7) business days of confirmation of the counterfeit status. The Contractor shall flow down a requirement for similar notification from Subcontractors to the Contractor. The Contractor shall initiate and submit an alert to the GIDEP within 30 calendar days of knowledge of the counterfeit part or material.

Counterfeit parts and material are defined by SAE AS5553 and SAE AS6174 may be electronic or mechanical in nature. Counterfeit electronic parts may typically be used parts which have been refurbished and represented as new. Commonly counterfeited electronic parts include parts such as microcontrollers or specially screened devices, or common parts, which have several pin-compatible versions from multiple manufacturers, such as memory devices and operational amplifiers. Counterfeit mechanical parts are typically improperly made, marked, or treated products. Examples are improper anodization or heat treatments, falsified data, mismarked parts sold as a higher grade steel, or used/fake parts such as valves or circuit breakers.

An authorized supplier is a supplier authorized by the original component manufacturer to buy parts or materials directly from the manufacturer. Parts provided from authorized suppliers typically have never left the manufacturer's authorized supply chain, and are accompanied by full manufacturer support and warranty.

3.14 REVIEWS

3.14.1 Make or Buy Program

New or alternate sources for production shall be required to qualify to the requirements of this contract and the TDP.

3.14.2 Make or Buy Program Changes

The Contractor shall notify the Government of Make-or Buy program changes in accordance with FAR 52.215-9, Changes or Additions to Make-or-Buy Program. The Government shall have the right to request, and receive from the Contractor, the detailed Make-or-Buy documentation during contract performance.

3.14.3 Program/Production Reviews

The Contractor shall hold quarterly program reviews commencing three (3) months after contract award, and every three months thereafter. The program reviews shall cover progress, schedules and problems. These reviews will be used to assess the Contractor's progress, risk in meeting contractual requirements and status. During these reviews, the Contractor shall provide, at a minimum, overall comprehensive status of:

- Production, procurement and material issues;
- Subcontractor and Vendor procurement
- Integrated Master Schedule including the Critical Path
- Make-or-Buy Program/Changes
- Risk Analysis Activities
- Quality Program Status
- Defect Reduction Program Status
- Safety Program Status
- Configuration Management Activities
- Government Furnished Equipment/Property Status and Issues
- Procurement Status

  - Purchase Order history
  - Expected delivery schedules
- Manufacturing Planning;
- Manufacturing Schedules;
- Production Processing & Fabrication;
- Calibration, Tool and Test Equipment;
- Non-Conforming Material Control;
- Completed Item Inspection and Test;
- Handling, Storage and Delivery
- Equipment and Data Item Delivery Schedule
- Final System Test(s)
- Cost Saving Initiatives
- Other Issues Identified by the Technical POC

The Contractor shall provide production plan to include all pertinent information as detailed above. The Contractor shall demonstrate the ability to manufacture and test the VLS Deck and Hatch Assy by providing various documents to include, but not be limited to, Manufacturing, Testing, Quality Assurance, Configuration Management, Calibration, and Supplier Management Plans. Policies, procedures, processes & plans demonstrating production processing & fabrication capability, facilities & equipment capacity, production scheduling, manpower resources & training, supplier management, parts & material procurement and shop loading will also be required. On-Site examination of the plans/documents described above, as well as examination of the manufacturing equipment and facility, review of support functions/groups, and interviews with project personnel, are all integral parts of the PRR. Processes must be able to produce repeatable conforming hardware within time constraints of the contract. During the PRR, the Contractor shall be prepared to discuss manufacturing & procurement planning status, issues, problems, risks and schedules associated with the production program, and provide requested supporting objective evidence.

A successful review is predicated on the Government's determination that there are no unacceptable risks, performance & quality requirements are fully met in the production configuration, and production capability demonstrates a satisfactory basis for proceeding into production. The Government in its sole discretion will determine whether the PRRs are successful.

The Integrated Master Schedule (IMS) shall be completed to Tier 1 in accordance with CDRL A006. In the case of any conflict between this SOW and CDRL A006, this SOW takes precedence. Tier 1 is defined as a major subcontractor.

For planning purposes, the program review location will be at the Contractor's facility.

The Contractor shall also conduct management meetings regarding delivery on a regular basis (to be mutually agreed upon between the Government and Contractor) between the program reviews. The meetings will be conducted via teleconference or at the Contractor's facility.

- Meetings may be held bi-weekly via teleconference

- Meeting materials shall be made available 24 hours prior to meeting start and distributed via email
- Bi-weekly review shall include updated to schedules for VLS Deck and Hatch Assy production status with actual start and completion dates for each major step and updated projections with detailed explanation if schedule slips
- Bi-weekly review shall include any special topics as mutually agreed upon between the Government and Contractor

The Contractor shall provide presentation material in accordance with CDRL A006. The Government has the right to hold additional informal reviews at mutually convenient times to follow progress and problems, which may exist.

3.14.4 Defect Reduction

The Contractor shall establish a statistically-based process control and defect reduction program to control and improve product quality. The Contractor shall present the progress of the process control and defect reduction program at the program reviews and/or periodic production reviews of SOW paragraph 3.14.3.

3.14.5 Program Management

The Contractor shall follow its internal Program Management Plan in the production of VLS Deck and Hatch Assy shall submit Progress, Status and Management Reports in accordance with CDRL A005.

3.15 Manufacturing Planning

The Contractor shall update their Manufacturing Plan to include VLS Deck and Hatch Assy being produced by the Contractor under this contract. The Contractor shall develop manufacturing and test procedures. The Contractor shall identify key production processes and shall establish process capability and process control requirements for each. The Contractor shall submit Progress, Status, and Management reports to cover progress, schedule performance, technical and quality problems and other issues as may be appropriate to facilitate program management by the Government in accordance with CDRL A005.

3.16 Risk Assessment and Management

The Contractor shall perform risk assessment and management. The Contractor shall use its own established Risk Management Program; however, the Risk Management Program shall at a minimum contain the following:

Risk management shall be a continuous process, analyzing program risks and individual mitigation plans on a sufficiently frequent basis to be proactive in preventing major Program problems. Major problems include, but are not limited to, not meeting delivery dates and nonconformance to specifications and drawings. The analysis shall identify the risks associated with each area, assess the probability of occurrence, identify the impact of each risk on the overall program, and develop and implement risk mitigation plans for identified risks. The risk analysis shall use metrics from the Contractor's Risk Management Program. Risk analysis status and metrics shall be reported in the Contractor's Progress, Status and Management Report in accordance with CDRL A005 and presented during Program and Technical Reviews.

3.17 Right of Access

The Government will have right of access to all facilities in which storage of raw materials, purchased parts, either finished or unfinished, and finished equipment may be established. Right of access shall include all areas in which VLS Deck and Hatch Assy are stored, manufactured, processed, aligned, assembled, and tested. Right of Access shall include access to records and documentation concerning the VLS Deck and Hatch Assy. Access includes specifications, purchase orders and subcontract documentation, receiving, inspection and test records for such purchase parts. Right of access shall include access at all tiers to vendors, Subcontractors and suppliers in accordance with FAR 46.405 and FAR clauses 52.245-1(g) and FAR 52.246-1.

The Government may perform any necessary inspections, verifications and evaluations to ascertain conformance to requirements, and the adequacy of the implementing procedures.

3.18 Subcontractor and Vendor Management

The Contractor shall identify and monitor Subcontractor and vendor technical, quality, schedule, and milestone achievement on a continuing basis, according to the Contractor's established subcontract and vendor management techniques. Notwithstanding the above, at a minimum the Contractor's subcontract and vendor management system shall comply with ANSI/ISO/ASQC Q9001 provisions for purchasing and supplier management. The Contractor shall:

- Establish, document, and maintain a purchasing system and develop a self-assessment program to ensure adequate controls;
- Acquire quality products at fair and reasonable prices, using best in class commercial purchasing practices and procedures and ensure fair and open competition;
- Only purchase components (consistent with section 13.3 of this SOW) from OEM, OCM, authorized suppliers or franchised distributors. (The terms OEM, OCM, authorized suppliers, and franchised distributors are each defined in SAE AS5553)
- Ensure, at a minimum, that procurement practices and processes to purchase and install components from OEM, OCM, authorized suppliers or franchised distributors are flowed down to subcontractors and suppliers at all tiers;
- Audit subcontracts as needed;
- Conduct review and periodic appraisal of the subcontractor's and vendor purchasing system and its self-assessment reports. The periodicity shall be at least annually. The Government shall have the right to examine and/or participate the reviews and appraisals' information; and
- Notify the Government prior to changes in Subcontractor status.

3.19 Special Tooling and Equipment Report

The Contractor shall design, fabricate or procure all special tooling, test equipment and gauges required to manufacture, assemble, test and inspect VLS Deck and Hatch Assy.

3.20 Contractor Storage of Equipment

Prior to and following the DD Form 250 acceptance and/or Program Quality Acceptance (PQA) signature, the Contractor shall provide material, services and facilities to store, preserve and protect any equipment completed during the manufacturing process until the specified contract delivery date.

3.21 QUALITY CONTROL AND ASSURANCE

3.21.1 Quality Control Program Plan

The Contractor shall submit a Quality Control Program Plan in accordance with CDRL A007. This document shall address all aspects of manufacturing, testing, and inspection as specified in the TDP, SOW, and contract. The Quality Control Program Plan should also address other aspects where quality control interface is appropriate, such as the quality group's input into purchasing decisions, and incoming receiving and inspection of material and components. The Contractor's Quality Control Program Plan must be approved by the Government before fabrication or procurement of any product for eventual delivery. Acceptance of the Contractor's quality control program shall not in any way relieve the Contractor of their responsibility for compliance with all contract requirements. All manufacturing, assembly, inspection, and testing shall adhere to the requirements of the QA plan. If any changes are made to the Quality Control Program Plan during the course of this contract, the Contractor shall notify the Government in writing of these changes.

3.21.2 Quality System and Quality Control Processes

The Contractor shall maintain a quality system, associated certifications or compliances, and test/inspection results. The Contractor shall maintain quality and process controls that shall be used to ensure that the manufactured products shall be in compliance with the applicable drawings, specifications, and this SOW. The Contractor shall maintain documentation and processes that shall be used to identify, record, and disposition non-conforming material, in-process rejects/reworks and characteristic discrepancies. This documentation shall be kept by the Contractor until further notification by the Government.

The quality management system procedures, planning, and all other documentation or data that comprise the quality management system shall be submitted to the Government for review in accordance with CDRL A007. Existing quality documents that meet the requirements of this contract may continue to be used. The Government may perform any necessary inspections, verifications, and evaluations to ascertain conformance to requirements and the adequacy of the implementing procedures. The Contractor's quality management system shall conform to supplement quality requirements imposed by this contract. The Government reserves the right to disapprove the quality management system or portions thereof when it fails to meet the contractual requirements.

3.21.3 Quality Assurance

The Contractor's quality assurance program shall conform to the Contractor's Quality Assurance Program Plan and shall be in conformity with either ANSI/ASQC Q9001, or ISO 9001 for production; by using ANSI/ASQC Q9001 as a guide. This applies to items manufactured or performed by the Contractor at the Contractor's plant or all outsourced facilities. The Contractor shall flow down to the Subcontractors and vendors the quality requirements through subcontracts, purchase orders, or other contractual purchasing documentation by using ANSI/ASQC Q9001 as a guide. The quality requirements include ANSI/ASQC Q9001 or ISO 9001, and relevant SOW and TDP requirements. The Subcontractors and vendors shall provide Certificates of Conformance that the flowed down requirements are fulfilled.

The Contractor shall provide for appropriate quality assurance review of manufacturing process plans and workmanship standards to ensure compliance to the quality system and standards.

The Contractor's Quality Assurance Program Plan shall be reviewed by the Government and must be accepted before fabrication or procurement of any product for eventual delivery. Acceptance of the Contractor's quality control program shall not in any way relieve the Contractor of their responsibility for compliance with all contract requirements. If any non-editorial changes are made to the Quality Assurance Program Plan during this contract, the Contractor shall notify the Government in writing of these changes. The Government reserves the right to disapprove the Quality Assurance Program Plan when it fails to meet the contractual requirements. The Contractor shall provide Quality Assurance Program Status describing the Contractor's activity as part of Status Report (Quality Assurance Program) in accordance with CDRL A007.

3.21.4 Manufacturing Certifications and Traceability

The Contractor shall submit a Certificate of Conformance certifying that the VLS Deck and Hatch Assy delivered meets the requirements of this contract.

The Contractor shall retain and provide upon request inspection and test results associated with certifying the units in accordance with CDRL A005.

3.21.5 Limited Shelf Life Components

The Contractor's quality management system shall ensure procured supplies that are subject to age deterioration, including epoxies and similar adhesives, are managed and controlled. Controls shall include prominent labeling and proactive measures to ensure expired material is not used in the manufacturing process. Controls shall allow complete traceability of limited shelf life components used in launcher products.

3.22 Government Inspection of Facilities

The Government may visit/inspect the plant or plants of the Contractor, or of any Subcontractors, engaged in the performance of this contract. If any examination/test is made by the Government on the premises of the Contractor/Subcontractor, the Contractor and any subcontractors shall provide all reasonable facilities and assistance for the safety and convenience of the Government inspectors in the performance of their duties. All examinations and tests by the Government will be performed in such a manner as will not unduly delay work.

3.23 Electrostatic Discharge Protection

The Contractor shall develop and maintain an electrostatic discharge protection/control program that complies with MIL-STD-1686C.

3.24 ENVIRONMENTAL COMPLIANCE AND REGULATIONS

3.24.1 Environmental Law and Regulations

The Contractor shall comply, and ensure that all Subcontractors comply, with all applicable environmental federal, state, and local laws and regulations and Navy policies, instructions and ISO 14000 Environmental Management Systems Standards.

3.24.2 Hazardous Waste and Material Control/Handling

The Contractor shall comply and ensure that all subcontractors comply, with all applicable environmental federal, state, local and Navy instructions for handling and control of hazardous waste.

3.24.3 Contractor use of Class I Ozone Depleting Substances

The TDP requirements notwithstanding, the Contractor shall not use 1,1,1-Trichloroethane or CFC-113 for cleaning or surface preparation. A substitute material that is not a Class I Ozone Depleting Substance shall be used to provide an equal or better level of cleanliness or surface preparation as provided by 1,1,1 Trichloroethane or CFC-113. The Contractor shall not use 1,1,1-Trichloroethane or CFC-113 as a solvent resistance test fluid. The Contractor shall notify the Government by written correspondence of its intent to perform material qualification tests which, by material specification, include 1,1,1-Trichloroethane or CFC-113 solvent resistance testing. The Contractor shall identify any detrimental impacts on the material qualification caused by not performing the 1,1,1-Trichloroethane or CFC-113 solvent resistance testing

3.24.4 Safety

The Contractor shall comply with the latest applicable federal and state laws, regulations and management plans and requirements regarding occupational safety and health.

3.25 Operational Security (OPSEC) Plan

Contractor will be developing, producing, analyzing, maintaining, transporting, storing, testing, or using critical information or indicators for this contract.

The contractor is required to institute OPSEC measures to protect the critical information and indicators of this program. The contractor shall comply with the Government Contracting Activity (GCA) or facilities OPSEC program instructions, guidance and contribute to organization-level OPSEC training and awareness programs while performing executions of this contract. Ensure you read and comply with NSWCPHDINST 3432.1C, updating NSWCPHDINST 3070.2 series for OPSEC requirements.

The contractor shall include OPSEC as part of its ongoing training and awareness program and complete all required activity OPSEC training. The contractor shall protect identified critical information, sensitive unclassified information and activities, which if divulged, could further compromise classified or sensitive information or operations, or degrade the planning and execution of operations performed in support of the mission.

3.26 International Traffic in Arms Regulations

MK 41 VLS as a missile launcher system is on U.S. Munitions List (USML). The contractor shall comply with ITAR requirements, and flow down this requirement to all suppliers.

(1) The contractor must be ITAR-registered with Directorate of Defense Trade Controls (DDTC).

(2) Technical data shall not be disclosed to foreign nationals without DDTC authorization.

(3) The contractor must implement access controls for all GFE technical data.

3.27 Dech and Hatch structure fabrication.

Deck and Hatch structure material is high yield strength steel. The contractor shall comply with the following requirements:

(1) The contractor shall have access to the gantry mill machine (capable to machine the large size Plenum and lower/upper Uptake structure) in a timely manner to meet contract schedule requirements.

(2) All materials shall be supplied with Certified Material Test Reports (CMTRs) showing chemical composition, mechanical properties, and heat treatment condition. Full traceability shall be maintained from raw material to the finished part.

(3) Welding procedure shall be qualified in accordance with NAVSEA S9074-AQ-GIB-010/248 (Procedures & Performance Qualification). Welders shall be qualified in accordance with NAVSEA S9074-AQ-GIB-010/248. Fabrication and inspection requirements

shall comply with NAVSEA S9074-AR-GIB-010/278. Filler metals shall meet MIL-E-22200/XX.

(4) Machining processes shall be controlled to prevent overheating, grinding burns or surface damage. Machine procedures shall include: cutting tool selection and parameters, coolant/lubricant usage, limits on climb milling, depth of cuts, and feeds to prevent work hardening.

(5) Post-machining inspection shall include 100% visual inspection, magnetic particle or liquid penetrant testing (PT) on critical surface.

(6) Deliverables shall include: (a) Certified Material Test Report (CMTRs); (b) Approved documents for welding; (c) Machining Process Specification (MPS); (d) Inspection and NDT reports; (e) Final inspection and acceptance records.

4.0 GENERAL DELIVERABLES

4.1 Technical Instructions/Deliverables

For any data generated under this contract, the contractor shall provide the Government unlimited rights or Government purpose rights. Anything less than that should be identified, disclosed in the appropriate TDP package and marked in each item. The Government reserves the right to use, reproduce and distribute such data, as it deems necessary.

The contractor shall be responsible for ensuring compliance with all copyright and trademark laws and appropriate marking of copyrighted and trademarked data, including obtaining permission for use and reproduction by the Government.

CDRL	Title	PARAGRAPH
A001	KEY EVENT SCHEDULE	3.2
A002	INSPECTION REPORT	3.2
A003	CONFIGURATION STATUS ACCOUNTING INFORMATION	3.4
A004	AS BUILT CONFIGURATION LIST-COMMON (ABCL-C)	3.4 and 3.10.1
A005	TECHNICAL REPORT STUDY/SERVICES	3.10.5, 3.14.3, 3.15, 3.16, 3.21.2
A006	CONTRACTOR'S PROGRESS, STATUS AND MANAGEMENT	3.14.1
A007	QUALITY CONTROL PROGRAM PLAN	3.21.1, 3.21.3, 3.21.4

ACRONYMS

The following acronyms appear in this Solicitation. This is not an exhaustive list.

ACRONYM	DEFINITION
CCB	Configuration Change Board

CDRL	Contract Discrepancy Report
DD	Dahlgren
ECP	Engineering Change Proposal
FACAR	Failure/Defect Reporting and Corrective Action System
FAI	First Article Inspection
GIDEP	Government-Industry Data Exchange Program
GFP	Government Furnished Property
NOR	Notice of Revision
NSWC	Naval Surface Warfare Center
OCM	Original Component Manufacturers
OEM	Original Equipment Manufacturer
OPSEC	Operation Security (OPSEC) Plan
PCA	Physical Configuration Audit
PHD	Port Hueneme Division
POC	Point of Contact
PRR	Production Readiness Review
QA	Quality Assurance
SOW	Statement of Work
TDP	Technical Data Package
VLS	Vertical Launching System